Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of $\begin{vmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix} = 0$, when m is equal to

(A) 6

(B) 5

(C) 4

(D) 1

Q.2

If $a > 0,\ b > 0, c > 0$ are respectively the pth, qth, rth terms of G.P., then the value of the determinant $\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix}$ is

(A) 0

(B) 1

(C) -1

(D) None of these

Q.3

If x, y, z are complex numbers, and $\Delta = \begin{vmatrix} 0 & -y & -z \\ \bar{y} & 0 & -x \\ \bar{z} & \bar{x} & 0 \end{vmatrix}$ then Δ is

(A) purely real

(B) purely imaginary

(C) complex

(D) 0

Q.4

@IITJEE_Advanced

If $a_1, a_2, a_3, \ldots\ldots\ldots\ldots\ldots\ldots$ are positive numbers in G.P. then the value of

$$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+1} & \log a_{n+2} & \log a_{n+3} \\ \log a_{n+2} & \log a_{n+3} & \log a_{n+4} \end{vmatrix}$$

(A) 1

(B) 4

(C) 3

(D) 0

Q.5

If $1_r^2 + m_r^2 + n_r^2 = 1; r = 1, 2, 3$ and
$l_r l_s + m_r m_s + n_r n_s = 0; r \neq s, r = 1, 2, 3; s = 1, 2, 3$
then the value of $\begin{vmatrix} 1_1 & m_1 & n_1 \\ 1_2 & m_2 & n_2 \\ 1_3 & m_3 & n_3 \end{vmatrix}$ is

(A) 0

(B) ± 1

(C) 2

(D) None of these

Q.6

If ω is a complex cube root of unity, then value of

$$\Delta = \begin{vmatrix} a_1 + b_1\omega & a_1\omega^2 + b_1 & c_1 + b_1\bar{\omega} \\ a_2 + b_2\omega & a_2\omega^2 + b_2 & c_2 + b_2\bar{\omega} \\ a_3 + b_3\omega & a_3\omega^2 + b_3 & c_3 + b_3\bar{\omega} \end{vmatrix}$$ is

(A) 0

(B) -1

(C) 2

(D) None of these

Q.7

If $a_r = (\cos 2r\pi + i \sin 2r\pi)^{1/9}$, then the value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is

(A) 1

@IITJEE_Advanced

(B) -1

(C) 0

(D) None of these

Q.8

If $\begin{vmatrix} \mathrm{a} & \cot \mathrm{A}/2 & \lambda \\ \mathrm{b} & \cot \mathrm{B}/2 & \mu \\ \mathrm{c} & \cot \mathrm{C}/2 & \gamma \end{vmatrix} = 0$, where $\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{A}, \mathrm{B}$, and C are elements of a triangle ABC with usual meaning.

Then, the value of a $(\mu - \gamma) + \mathrm{b}(\gamma - \lambda) + \mathrm{c}(\lambda - \mu) = 0$ is

(A) 0

(B) abc

(C) $\mathrm{ab} + \mathrm{bc} + \mathrm{ca}$

(D) $2\mathrm{abc}$

Q.9

The determinant

$$\begin{vmatrix} a+b+c & a+b & a \\ 4a+3b+2c & 3a+2b & 2a \\ 10a+6b+3c & 6a+3b & 3a \end{vmatrix}$$

is independent of which one of the following?

(A) a and b

(B) b and c

(C) a and c

(D) All of these

Q.10

If $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are in GP, then what is the value of

$$\begin{vmatrix} a & b & a+b \\ b & c & b+c \\ a+b & b+c & 0 \end{vmatrix} ?$$

(A) 0

(B) 1

(C) -1

(D) None of these

@IITJEE_Advanced

Q.11

If $A_1B_1C_1$, $A_2B_2C_2$ and $A_3B_3C_3$ are three digit numbers, each of which is divisible by k, then $\Delta = \begin{vmatrix} \mathrm{A}_1 & \mathrm{B}_1 & \mathrm{C}_1 \\ \mathrm{A}_2 & \mathrm{B}_2 & \mathrm{C}_2 \\ \mathrm{A}_3 & \mathrm{B}_3 & \mathrm{C}_3 \end{vmatrix}$ is

(A) divisible by k

(B) divisible by k^2

(C) divisible by k^3

(D) None of these

Q.12

$\mathrm{If} f(x) = \begin{vmatrix} 2^{-x} & e^{x\log_e 2} & x^2 \\ 2^{-3x} & e^{3x\log_e 2} & x^4 \\ 2^{-5x} & e^{5x\log_e 2} & 1 \end{vmatrix}$, then

(A) $f(x) + f(-x) = 0$

(B) $f(x) - f(-x) = 0$

(C) $f(x) + f(-x) = 2$

(D) None of these

Q.13

If in a triangle ABC, $\begin{vmatrix} 1 & \sin A & \sin^2 A \\ 1 & \sin B & \sin^2 B \\ 1 & \sin C & \sin^2 C \end{vmatrix} = 0$ then the triangle is

(A) equilateral or isosceles

(B) equilateral or right-angled

(C) right angled or isosceles

(D) None of these

Q.14

If A, B, and C are the angles of a triangle and $\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A+\sin^2 A & \sin B+\sin^2 B & \sin C+\sin^2 C \end{vmatrix} = 0$ then the triangle must be

(A) Isosceles

@IITJEE_Advanced

(B) Equilateral

(C) Right-angled

(D) None of these

Q.15

$A = \begin{vmatrix} 2a & 3r & x \\ 4b & 6s & 2y \\ -2c & -3t & -z \end{vmatrix} = \lambda \begin{vmatrix} a & r & x \\ b & s & y \\ c & t & z \end{vmatrix}$, then what is the value of λ ?

(A) 12

(B) -12

(C) 7

(D) -7

Q.16

The value of the determinant $\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix}$ is :

(A) $2(a+b+c)$

(B) $2(a+b+c)^2$

(C) $2(\mathrm{a}+\mathrm{b}+\mathrm{c})^3$

(D) $(2a+2b+2c)^3$

Q.17

If $\begin{matrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{matrix} = 0$ where $a, b, c \neq 1$, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to

(A) 0

(B) 1

(C) 2

(D) -2

Q.18

The value of $\begin{vmatrix} 1 & 2 & 3 \\ 3 & 5 & 7 \\ 8 & 14 & 20 \end{vmatrix}$ is equal to:

@IITJEE_Advanced

(A) 20

(B) 200

(C) 0

(D) 250

Q.19

If $1, \omega, \omega^2$ are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$ is equal to

(A) 0

(B) 1

(C) ω

(D) ω^2

Q.20

The value of $\begin{vmatrix} b+c & a+b & a \\ c+a & b+c & b \\ a+b & c+a & c \end{vmatrix}$ is :

(A) $\mathrm{a}+\mathbf{b}+\mathbf{c}$

(B) $(a+b+c)^2$

(C) $(a+b+c)^3$

(D) $a^3+b^3+c^3-3abc$

Answers & Solutions

Q.1 Answer:

5

Solution:

$$\begin{vmatrix} {}^{10}\mathrm{C}_4 & {}^{10}\mathrm{C}_5 & {}^{11}\mathrm{C}_{\mathrm{m}} \\ {}^{11}\mathrm{C}_6 & {}^{11}\mathrm{C}_7 & {}^{12}\mathrm{C}_{\mathrm{m}+2} \\ {}^{12}\mathrm{C}_8 & {}^{12}\mathrm{C}_9 & {}^{13}\mathrm{C}_{\mathrm{m}+4} \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} {}^{10}\mathrm{C}_4 + {}^{10}\mathrm{C}_5 & {}^{10}\mathrm{C}_5 & {}^{11}\mathrm{C}_{\mathrm{m}} \\ {}^{11}\mathrm{C}_6 + {}^{11}\mathrm{C}_7 & {}^{11}\mathrm{C}_7 & {}^{12}\mathrm{C}_{\mathrm{m}+2} \\ {}^{12}\mathrm{C}_8 + {}^{12}\mathrm{C}_9 & {}^{12}\mathrm{C}_9 & {}^{13}\mathrm{C}_{\mathrm{m}+4} \end{vmatrix} = 0$$

$(\mathrm{R}_1 \to \mathrm{R}_1 + \mathrm{R}_2)$

$$\Rightarrow \begin{vmatrix} {}^{11}\mathrm{C}_5 & {}^{10}\mathrm{C}_5 & {}^{11}\mathrm{C}_{\mathrm{m}} \\ {}^{12}\mathrm{C}_7 & {}^{11}\mathrm{C}_7 & {}^{12}\mathrm{C}_{\mathrm{m}+2} \\ {}^{13}\mathrm{C}_9 & {}^{12}\mathrm{C}_9 & {}^{13}\mathrm{C}_{\mathrm{m}+4} \end{vmatrix} = 0$$

@IITJEE_Advanced

It means either two rows or two columns are identical $\therefore {}^{11}C_5 = {}^{11}C_m, {}^{12}C_7 = {}^{12}C_{m+2}, {}^{13}C_4 = {}^{13}C_{m+4} \quad \Rightarrow m = 5$

Q.2 Answer:

0

Solution:

Let A **be the 1 st term and** R **the common ratio of G.P., then;** $a = T_p = AR^{p-1} \quad \therefore \log a = \log A + (p-1)\log R$

Similarly, $\log b = \log A + (q-1)\log R$

and $\log c = \log A + (r-1)\log R$

$$\therefore \Delta = \begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log A + (q-1)\log R & q & 1 \\ \log A + (r-1)\log R & r & 1 \end{vmatrix}$$

Split into two determinants and in the first take logA common and in the second take $\log R$ **common**

$$\Delta = \log A \begin{vmatrix} 1 & p & 1 \\ 1 & q & 1 \\ 1 & r & 1 \end{vmatrix} + \log R \begin{vmatrix} p-1 & p & 1 \\ q-1 & q & 1 \\ r-1 & r & 1 \end{vmatrix}$$

Apply $C_1 \to C_1 - C_2 + C_3$ **in the second**

$$\Delta = 0 + \log R \begin{vmatrix} 0 & p & 1 \\ 0 & q & 1 \\ 0 & r & 1 \end{vmatrix} = 0$$

Q.3 Answer:

purely imaginary

Solution:

We have $\bar{\Delta} = \begin{vmatrix} 0 & -\bar{y} & -\bar{z} \\ y & 0 & -\bar{x} \\ z & x & 0 \end{vmatrix} = \begin{vmatrix} 0 & y & z \\ -\bar{y} & 0 & x \\ -\bar{z} & -\bar{x} & 0 \end{vmatrix}$

[Interchanging rows and columns] $= (-1)^3 \begin{vmatrix} 0 & -y & -z \\ \bar{y} & 0 & -x \\ \bar{z} & \bar{x} & 0 \end{vmatrix}$

[Taking-1 common from each row] $= -\Delta$

$\bar{\Delta} + \Delta = 0 \Rightarrow 2\,\mathrm{Re}(\Delta) = 0$

Δ is purely imaginary.

Q.4 Answer:

0

Solution:

@IITJEE_Advanced

(d) If the G.P be $a, ar, ar^2, \ldots$ then $a_n = ar^{n-1}$ $D =$

$$\begin{vmatrix} \log a + (n-1)\log r & \log a + n\log r & \log a + (n+1)\log r \\ \log a + n\log r & \log a + (n+1)\log r & \log a + (n+2)\log r \\ \log a + (n+1)\log r & \log a + (n+2)\log r & \log a + (n+3)\log r \end{vmatrix}$$

$R_3 \to R_3 - R_2$ and $R_2 \to R_2 - R_1$ gives,

$$= \begin{vmatrix} \log a + (n-1)\log r & \log a + n\log r & \log a + (n+1)\log r \\ \log r & \log r & \log r \\ \log r & \log r & \log r \end{vmatrix}$$

$= 0$, since R_2 and R_3 are identical.

Q.5 Answer:

± 1

Solution:

$$D^2 = \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}$$

$$= \begin{vmatrix} l_1^2 + m_1^2 + m_1^2 & l_2 l_1 + m_2 m_1 + n_2 n_1 & l_1 l_3 + m_1 m_3 + n_1 n_3 \\ l_2 l_1 + m_2 m_1 + n_2 n_1 & l_2^2 + m_2^2 + n_2^2 & l_2 l_3 + m_2 m_3 + n_2 n_3 \\ l_1 l_3 + m_1 m_3 + n_1 n_3 & l_2 l_3 + m_2 m_3 + n_2 n_3 & l_3^2 + m_3^2 + n_3^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \quad \Rightarrow D = \pm 1$$

Q.6 Answer:

0

Solution:

$$\Delta = \begin{vmatrix} a_1 + b_1\omega & a_1\omega^2 + b_1 & c_1 + b_1\bar{\omega} \\ a_2 + b_2\omega & a_2\omega^2 + b_2 & c_2 + b_2\bar{\omega} \\ a_3 + b_3\omega & a_3\omega^2 + b_3 & c_3 + b_3\bar{\omega} \end{vmatrix}$$

Using $C_2 \to \omega C_2$

We have $\Delta = \frac{1}{\omega} \begin{vmatrix} a_1 + b_1\omega & a_1\omega^3 + b_1\omega & c_1 + b_1\bar{\omega} \\ a_2 + b_2\omega & a_2\omega^3 + b_2\omega & c_2 + b_2\bar{\omega} \\ a_3 + b_3\omega & a_3\omega^2 + b_3\omega & c_3 + b_3\bar{\omega} \end{vmatrix}$

$$= \frac{1}{\omega} \begin{vmatrix} a_1 + b_1\omega & a_1 + b_1\omega & c_1 + b_1\bar{\omega} \\ a_2 + b_2\omega & a_2 + b_2\omega & c_2 + b_2\bar{\omega} \\ a_3 + b_3\omega & a_3 + b_3\omega & c_3 + b_3\bar{\omega} \end{vmatrix} = 0$$

Q.7 Answer:

0

Solution:

@IITJEE_Advanced

$\cos\alpha + i\sin\alpha = \mathrm{e}^{\alpha}$ **... (Euler's representation of complex number)**

$a_{\mathrm{r}} = (\cos 2r\pi + i\sin 2r\pi)^{1/9}$

$\therefore a_{\mathrm{r}} = e^{\frac{i2r\pi}{9}}$

$\therefore a_{\mathrm{r}} = e^{\mathrm{r}\theta} \quad \ldots \left[\theta = \frac{i2\pi}{9}\right]$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$$

$$= \begin{vmatrix} e^{\theta} & e^{2\theta} & e^{3\theta} \\ e^{4\theta} & e^{5\theta} & e^{6\theta} \\ e^{7\theta} & e^{8\theta} & e^{9\theta} \end{vmatrix}$$

$$= \left(e^{3\theta}\right) \begin{vmatrix} e^{\theta} & e^{2\theta} & e^{3\theta} \\ e^{\theta} & e^{2\theta} & e^{3\theta} \\ e^{7\theta} & e^{8\theta} & e^{9\theta} \end{vmatrix}$$

$= 0$

Hence, the answer is option (C).

Q.8 Answer:

0

Solution:

@IITJEE_Advanced

$$\text{(a) Given, } \begin{vmatrix} a & \cot A/2 & \lambda \\ b & \cot B/2 & \mu \\ c & \cot c/2 & \gamma \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} \mathrm{a} & \frac{\mathrm{s(s-a)}}{\Delta} & \lambda \\ \mathrm{b} & \frac{\mathrm{s(s-b)}}{\Delta} & \mu \\ \mathrm{c} & \frac{\mathrm{s(s-c)}}{\Delta} & \gamma \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{r}\begin{vmatrix} a & s-a & \lambda \\ b & s-b & \mu \\ c & s-c & \gamma \end{vmatrix} = 0$$

Apply $\mathrm{C_2 \to C_2 + C_1}$

$$\Rightarrow \frac{1}{\mathrm{r}}\begin{vmatrix} \mathrm{a} & \mathrm{s} & \lambda \\ \mathrm{b} & \mathrm{s} & \mu \\ \mathrm{c} & \mathrm{s} & \gamma \end{vmatrix} = 0, \text{ where } \mathrm{r} = \frac{\Delta}{\mathrm{s}}$$

$$\Rightarrow \frac{\Delta}{r^2}\begin{vmatrix} a & 1 & \lambda \\ b & 1 & \mu \\ c & 1 & \gamma \end{vmatrix} = 0$$

Apply $\mathrm{R_1 \to R_1 - R_2, R_2 \to R_2 - R_3}$

$$\Rightarrow \frac{\Delta}{r^2}\begin{vmatrix} a-b & 0 & \lambda-\mu \\ b-c & 0 & \mu-\gamma \\ c & 1 & \gamma \end{vmatrix} = 0$$

$$\Rightarrow \frac{\Delta}{r^2}[(b-c)(\lambda-\mu)-(\mu-\gamma)(a-b)] = 0$$

$$\Rightarrow \quad \mathrm{b}(\lambda-\mu)-\mathrm{c}(\lambda-\mu)-\mathrm{a}(\mu-\gamma)+\mathrm{b}(\mu-\gamma) = 0$$

$$\Rightarrow \quad -\mathrm{a}(\mu-\gamma)+\mathrm{b}(\lambda-\mu+\mu-\gamma)-\mathrm{c}(\lambda-\mu) = 0$$

$$\Rightarrow \quad -\mathrm{a}(\mu-\gamma)+\mathrm{b}(\lambda-\gamma)-\mathrm{c}(\lambda-\mu) = 0$$

$$\Rightarrow \quad \mathrm{a}(\mu-\gamma)+\mathrm{b}(\gamma-\lambda)+\mathrm{c}(\lambda-\mu) = 0$$

Q.9 Answer:

b and c

Solution:

Let $D = \begin{vmatrix} a+b+c & a+b & a \\ 4a+3b+2c & 3a+2b & 2a \\ 10a+6b+3c & 6a+3b & 3a \end{vmatrix}$

$$\Rightarrow \quad D = \begin{vmatrix} a+b+c & a+b & a \\ 4a+3b+2c & 3a+2b & 2a \\ 10a+6b+3c & 6a+3b & 3a \end{vmatrix}$$

By $\mathrm{R_2 \to R_2 - 2R_1}$ **and** $\mathrm{R_3 \to R_3 - 3R_1}$**, we get**

$$\Rightarrow \begin{vmatrix} \mathrm{c} & \mathrm{a+b} & \mathrm{a} \\ \mathrm{a+b} & \mathrm{a} & 0 \\ \mathrm{4a+3\,b} & \mathrm{3a} & 0 \end{vmatrix}$$

Again by $\mathrm{R_3 \to R_3 - 3R_1}$**, we get :**

@IITJEE_Advanced

$$D = \begin{vmatrix} a+b+c & a+b & a \\ a+b & a & 0 \\ a & 0 & 0 \end{vmatrix}$$

$= a\{0 \cdot (a+b) - a \cdot a\}$

$= -a^3$ **which is independent of** b **and** c

Q.10 Answer:

0

Solution:

Since, a, b, c **are in GP.** $\Rightarrow b^2 = ac$

Explanding the determinant we get, $\begin{vmatrix} a & b & a+b \\ b & c & b+c \\ a+b & b+c & 0 \end{vmatrix}$

$= a\begin{vmatrix} c & b+c \\ b+c & 0 \end{vmatrix} - b\begin{vmatrix} b & b+c \\ a+b & 0 \end{vmatrix}$

$+(a+b)\begin{vmatrix} b & c \\ a+b & b+c \end{vmatrix}$

$= -a(b+c)^2 + b(a+b)(b+c) + (a+b)$

$\left(b^2 + bc - ac - bc\right)$

$= -a\left(b^2 + c^2 + 2bc\right) + b\left(ab + ac + b^2 + bc\right)$

$= -ab^2 - ac^2 - 2abc + ab^2 + 2abc + b^2c$

$\left(\because b^2 = ac\right)$

$= -ac^2 + b^2c = -ac^2 +$ **ac.c.** $= -ac^2 + ac^2 = 0$

Q.11 Answer:

divisible by k

Solution:

Since, $A_1B_1C_1$, $A_2B_2C_2$ **and** $A_3B_3C_3$ **are divisible byk, therefore;** $100\ A_1 + 10\ B_1 + C_1 = n_1k$

$100\ A_2 + 10\ B_2 + C_2 = n_2k; 100\ A_3 + 10\ B_3 + C_3 = n_3k$

(where n_1, n_2, n_3 **are integers)**

Now, $\Delta = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$

$= \begin{vmatrix} A_1 & B_1 & 100\ A_1 + 10\ B_1 + C_1 \\ A_2 & B_2 & 100\ A_2 + 10\ B_2 + C_2 \\ A_3 & B_3 & 100\ A_3 + 10\ B_2 + C_3 \end{vmatrix}$

[**Applying** $C_3 \to C_3 + 10C_2 + 100C_1$]

@IITJEE_Advanced

$$= \begin{vmatrix} A_1 & B_1 & n_1 k \\ A_2 & B_2 & n_2 k \\ A_3 & B_3 & n_3 k \end{vmatrix} = k \begin{vmatrix} A_1 & B_1 & n_1 \\ A_2 & B_2 & n_2 \\ A_3 & B_3 & n_3 \end{vmatrix} = k\Delta_1$$

$\Rightarrow \Delta$ **is divisible by** k **[Since, elements of** Δ_1 **are integers** $\therefore \Delta_1$ **is an integer.]**

Q.12 Answer:

$f(x) + f(-x) = 0$

Solution:

$$f(x) = \begin{vmatrix} 2^{-x} & 2^{x} & x^2 \\ 2^{-3x} & 2^{3x} & x^4 \\ 2^{-5x} & 2^{5x} & 1 \end{vmatrix}$$

$$\therefore f(-x) = \begin{vmatrix} 2^{x} & 2^{-x} & x^2 \\ 2^{3x} & 2^{-3x} & x^4 \\ 2^{5x} & 2^{-5x} & 1 \end{vmatrix} = -f(x)$$

Q.13 Answer:

equilateral or isosceles

Solution:

$$\begin{vmatrix} 1 & \sin A & \sin^2 A \\ 1 & \sin B & \sin^2 B \\ 1 & \sin C & \sin^2 C \end{vmatrix} = 0$$

$\Rightarrow (\sin \mathrm{A} - \sin \mathrm{B})(\sin \mathrm{B} - \sin \mathrm{C})(\sin \mathrm{C} - \sin \mathrm{A}) = 0$

$\Rightarrow \sin \mathrm{A} = \sin \mathrm{B}$ **or** $\sin \mathrm{B} = \sin \mathrm{C}$ **or** $\sin \mathrm{C} = \sin \mathrm{A}$

$\therefore$ **atleast two of** $\mathrm{A}, \mathrm{B}, \mathrm{C}$ **are equal. Hence the triangle is isosceles or equilateral.**

Q.14 Answer:

Isosceles

Solution:

Using $C_2 \to C_2 - C_1$ **and** $C_3 \to C_3 - C_1$ **in the given determinant, we have** $\Delta = \begin{vmatrix} 1 & 0 & 0 \\ 1 + \sin A & \sin B - \sin A & \sin C - \sin A \\ \sin A + \sin^2 A & \sin^2 B - \sin^2 A & \sin^2 C - \sin^2 A \end{vmatrix}$

Now taking $\sin \mathrm{B} - \sin \mathrm{A}$ **common from** C_2 **and**

$\sin$

$\mathrm{C} - \sin \mathrm{A}$ **common from** C_3**, we have**

$\Delta = (\sin \mathrm{B} - \sin \mathrm{A})(\sin \mathrm{C} - \sin \mathrm{A})$

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 + \sin A & 1 & 1 \\ \sin A + \sin^2 A & \sin B + \sin A & \sin C + \sin A \end{vmatrix}$$

$= (\sin \mathrm{B} - \sin \mathrm{A})(\sin \mathrm{C} - \sin \mathrm{A})(\sin \mathrm{C} - \sin \mathrm{B})$

As the determinant is zero, we must have $\sin \mathrm{B} = \sin \mathrm{A}$ **or** $\sin \mathrm{A}$ **or** $\sin \mathrm{C} = \sin \mathrm{A}$ **or** $\sin \mathrm{C} =$

@IITJEE_Advanced

$\sin B$

that is, $B = A$ **or** $C = A$ **or** $C = B$.

In all three cases we will have an isosceles triangle.

Q.15 Answer:

-12

Solution:

Given, $\begin{vmatrix} 2a & 3r & x \\ 4b & 6s & 2y \\ -2c & -3t & -z \end{vmatrix} = \lambda \begin{vmatrix} a & r & x \\ b & s & y \\ c & t & z \end{vmatrix}$

Taking 2 common from C_1 **and 3 from** C_2 **in LHS**

$\therefore 2 \times 3 \begin{vmatrix} a & r & x \\ 2b & 2s & 2y \\ -c & -t & -z \end{vmatrix} = \lambda \begin{vmatrix} a & r & x \\ b & s & y \\ c & t & z \end{vmatrix}$

Taking 2 common from R_2 **and** -1 **from** R_3 **in LHS**

$\therefore -12 \begin{vmatrix} a & r & x \\ b & s & y \\ c & t & z \end{vmatrix} = \lambda \begin{vmatrix} a & r & x \\ b & s & y \\ c & t & z \end{vmatrix}$

$\Rightarrow \lambda = -12$

Q.16 Answer:

$2(a+b+c)^3$

Solution:

L.H.S. $= \begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$

$= \begin{vmatrix} 2(a+b+c) & a & b \\ 2(a+b+c) & b+c+2a & b \\ 2(a+b+c) & a & c+a+2b \end{vmatrix}$

$= 2(a+b+c) \begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$

Applying $R_1 \to R_1 - R_2$ **and** $R_2 \to R_2 - R_3$

$= 2(a+b+c) \begin{vmatrix} 0 & -(a+b+c) & 0 \\ 0 & (a+b+c) & -(a+b+c) \\ 1 & a & c+a+2b \end{vmatrix}$

$= 2(a+b+c+)^3 \begin{vmatrix} 0 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & a & c+a+2b \end{vmatrix} =$

$= 2(a+b+c)^3[1(1-0)] = 2(a+b+c)^3 =$ **RHS**

@IITJEE_Advanced

Q.17 Answer:

1

Solution:

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a-1 & 0 & 1 \\ 1-b & b-1 & 1 \\ 0 & 1-c & c \end{vmatrix}$$

$= 0$ **expand and divided by** $1-a1-b1-c$

$\Rightarrow \frac{c}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 0$

Q.18 Answer:

0

Solution:

Applying $C_2 - 2C_1$, **then** $C_3 - 3C_1$, **we have** $\Delta = \begin{vmatrix} 1 & 0 & 0 \\ 3 & -1 & -2 \\ 8 & -2 & -4 \end{vmatrix} = 0$

Q.19 Answer:

0

Solution:

$$\Delta = \begin{vmatrix} 1+m^n+\omega^{2n} & m^n & \omega^{2n} \\ 1+\omega^n+\omega^{2n} & m^{2n} & 1 \\ 1+\omega^n+\omega^2 n & 1 & 0^n \end{vmatrix}$$

$= 0$

Since, $1+\omega^n+\omega^{2n} = 0$, **if** n **is not a multiple of**

Therefore, the roots are identical.

Q.20 Answer:

$a^3+b^3+c^3-3abc$

Solution:

$$\Delta = (\mathrm{a}+\mathrm{b}+\mathrm{c}) \begin{vmatrix} 1 & a+b & a \\ 1 & b+c & b \\ 1 & c+a & c \end{vmatrix} \quad (c_2 \to C_1 + c_1)$$

$$= (a+b+c) \begin{vmatrix} 1 & a+b & a \\ 0 & c-a & b-a \\ 0 & c-b & c-a \end{vmatrix}$$

$$= (a+b+c)\left(c^2+a^2-2ca-bc+b^2+ac-ab\right)$$

$$= (a+b+c)\left(a^2+b^2+c^2-ab-bc-ca\right)$$

$$= a^3+b^3+c^3-3abc$$

@IITJEE_Advanced